Anna Tsvetkov

Brown University
Department of Philosophy
Department of Computer Science

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Areas of Interests

AOS: Philosophy of AI (including AI Ethics) · Philosophy of Mind · Philosophy of Cognitive Science
AOC: Philosophy of Language · Logic and Formal Methods

Education

PhD in Philosophy, Brown University 2018 - 2025
Dissertation: *Human-Centered Artificial Intelligence*
Committee: Adam Pautz (Chair), Chris Hill, Ellie Pavlick, David Chalmers (NYU)

ScM in Computer Science, Brown University 2023 - 2025
Thesis: *Can Large Language Models Represent Perceptual Reality?*
Advisor: Ellie Pavlick

BA in Philosophy, Binghamton University, *summa cum laude* 2014 - 2018

Publications

Articles

Murphy, R. A., Witnauer, J. E., Castiello, S., Tsvetkov, A., Li, A., Alcaide, D. M., & Miller, R. R. (2022). "More frequent, shorter trials enhance acquisition in a training session: There is a free lunch!" *Journal of Experimental Psychology: General*

Under Review

Tsvetkov, A. (2024). "Troubles with Spatial Functionalism"

In Progress

Tsvetkov, A. & Pavlick, E. (2024). "Can Large Language Models Represent Perceptual Reality?"
Tsvetkov, A. "Molyneux's Question and Multimodal Models"
Tsvetkov, A., Trisovic, A., Thompson, N. "A Large-Scale Analysis of Ethical Considerations in AI Foundation Models" (In progress with *MIT FutureTech.)*
Trisovic, A., Tsvetkov, A., Fogelson, A., Siva, J., Thompson, N. "Scientific Use of Foundation Models and Democratization of AI" (In progress with *MIT FutureTech.)*

Experience

MIT FutureTech
AI Researcher, Cambridge, MA Current

NSF AI Institute for Artificial Intelligence and Fundamental Interactions (IAIFA)
PhD Summer School Participant, Boston, MA Summer 2023, 2024

AI Safety Student Team at Harvard (AISST)
Technical AI Safety Fellow, Cambridge, MA Spring 2023

Presentations and Comments

(∗ = invited, otherwise peer reviewed.)

'Scientific Use of Foundation Models and Democratization of AI (with Ana Trisovic)"
MIT Initiative on the Digital Economy, MIT (May 2024) *

"Comments on Umrao Sethi (Brandeis University)"
New England Workshop on Mental Things, Rhode Island College (Nov. 2022) *

"Comments on Julia Minarik (UToronto)"
Athena in Action, Rutgers University (June 2022)

"Counting vs. Timing Intertrial Intervals During Conditioning"
International Conference on Comparative Cognition, Melbourne, FL (Apr. 2018)

Teaching

(* = an online course)

Brown University
Knowledge and Reality (Spring 2022), *Full Instructor*
Consciousness (Fall 2019, Fall 2021) *Teaching Assistant*
Philosophy of Mathematics (Fall 2020*) *Teaching Assistant*
Early Modern Philosophy (Spring 2020*) *Teaching Assistant*
Ancient Greek Philosophy (Fall 2023) *Teaching Assistant*
Ethical Themes in the Contemporary American Short Story (Spring 2024) *Teaching Assistant*
Place of Persons (Spring 2021*, Fall 2024) *Teaching Assistant*

Summer@Brown Pre-College Program
Philosophy and Psychology of Happiness (Summer 2019) *Teaching Assistant*

Graduate Coursework

(* = an audited course)

In Philosophy
Perception*
Mental Representation*
Philosophy of Science
Reductionism
Introspection
Inquiry
Kant: Critique of Pure Reason
Aristotle's Psychology
Disagreement
Moral Psychology
Independent Study (with Adam Pautz)
Independent Study (with Elizabeth Miller)

In Computer Science
Artificial Intelligence
Deep Learning
Computational Linguistics
Reading and Research I (with Ellie Pavlick)
Reading and Research II (with Ellie Pavlick)
Cybersecurity Law and Policy
Interpretability of Language Models

In Other Institutions
Hume's Treatise and Contemporary Themes in Humean Metaphysics (MIT)
The First-Person (Harvard)

Programming Competence

Languages Python
Libraries NumPy · Pytorch · TensorFlow · Transformers · matplotlib

Service

Member of Minorities and Philosophy (MAP) Chapter (2018 - 2024)
Member of the Climate Committee Working Group (2020 - 2022)
Co-organizer, Brown's Mark L. Shapiro Graduate Philosophy Conference (2022)
Co-organizer, Philosophy Department, Inclusive Teaching Workshop (2019)
Graduate Student Mentor, Brown AI Safety Team (BAIST) and MIT AI Alignment (MAIA) (2023-2024)

References

Adam Pautz
Professor of Philosophy
Brown University
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Ellie Pavlick
Briger Family Distinguished Associate Professor of Computer Science
Brown University
Email: `ellie_pavlick@brown.edu`

Ana Trisovic
AI Research Scientist
MIT FutureTech
Email: `ana_tris@mit.edu`

Dissertation Abstract

Ask a question to ChatGPT and there's a good chance you'll be impressed at what it says–unless it spits out some malarkey. Part of what is so puzzling about artificial intelligence (AI) models like ChatGPT is that no one knows precisely how they work. And without knowing how the models work, it's difficult to say what, if anything, they can teach us about ourselves. Of course, we don't *need* to know how something works to use it. I drive a car, I admit, without knowing the first thing about what goes on under the hood. But if we want to know what AI can teach us about big questions concerning the mind, language, and ethics my dissertation makes the argument that we had better get our hands dirty and look under the hood of AI models. (Can't say the same for my car.)

How we should approach this task is as much a philosophical question as it is a technical one. We can start big and ask, "Do AI models *really* understand questions about the world? Can ChatGPT have anything like human-level language comprehension, values, or cognition?". Or we can start small and ask, "Does this particular neuron in the model respond in predictable ways to *that* feature?" My dissertation is a collection of chapters that bridge philosophy and computer science to tackle both the big and the small questions. My dissertation develops a *human-centered artificial intelligence* research program that uses philosophy to guide carefully designed AI experiments to deepen our understanding of the mind and, in turn, uses those insights to create explainable and ethical AI. A key lesson emerging from the chapters that by designing experiments informed by philosophy and rolling up our sleeves to look "under the hood" of AI models, we can see what they represent—the good and the bad—and breathe new life into longstanding philosophical debates.

Chapter 1 examines whether large language models (LLMs) like ChatGPT can represent the world like we do. This chapter, based on a paper co-authored with Ellie Pavlick, uses probing techniques inspired by the human brain and a philosophically informed notion of representation to look under the hood of

large language models to understand how they represent perceptual reality. By examining the nature of representations in AI systems, we help philosophers connect AI with fundamental questions about the mind while also helping computer scientists situate their work within philosophical debates and find new directions for empirical research. We find that LLMs have specialized perceptual neurons—"color neurons" and "space neurons"—that form subnetworks that encode and process information about color and spatial relationships. We tweak these neurons in targeted ways to demonstrate that the neurons form causal subnetworks that actually drive perceptual processing—enough to trick the models into thinking red is green and left is right with our interventions. Our work suggests that LLMs can learn representations that reflect the structure of peceptual reality from text alone—*no senses required!* This research advances our ability to make meaningful comparisons between artificial and natural minds and has important ethical upshots for how we can better interpret AI systems to ensure they align with the positive aspects of human cognition.

Chapter 2 explores whether AI can be used to make progress on longstanding philosophical debates. This chapter revisits the famous philosophical thought experiment Molyneux's Question—which asks whether someone born blind could recognize shapes by sight if they suddenly gained vision—using AI. We connect language and vision models to simulate cross-modal learning, exploring whether a language model trained on text alone can, upon receiving visual input, recognize shapes it previously knew only through language. The experiment bridges AI and philosophy to provide insights into perception and learning transfer across modalities in both natural and artificial minds. We explain how multimodal models provide new ways to overcome the limitations of empirical studies that use human subjects who have undergone sight restoration to address Molyneux's Question. The reader who is skeptical of what AI can teach us about ourselves should treat this chapter as an invitation to see what can be gained when the rubber meets the road—when philosophy and AI work together to illuminate how the mind works.

Chapter 3 shifts gears from analyzing the inner workings of AI models to exploring how AI is used in the wild. As we integrate AI into high-stakes areas like healthcare, criminal justice, and social media, ensuring that these systems align with human values is more important than ever. This chapter, based on a paper written in collaboration with MIT's FutureTech Lab, examines the ethical considerations behind the development of foundation models—large-scale AI models like ChatGPT that serve as a foundation, or basis, for many downstream applications. We analyze how fairness, transparency, bias, safety, and other ethical principles are addressed in the development of foundation models. Our framework identifies key gaps in current practices and offers policy solutions to promote the development of ethical AI. This chapter highlights the importance of aligning AI development with human values and makes recommendations for the future trajectory of AI to ensure these technologies contribute positively to science and society at large.

A unifying thread running through the chapters is that bridging big questions with technical science requires grappling with the intricacies of just how a scientific approach to the philosophy of AI should proceed. My work is motivated by the ideal that the philosophy of AI can be *both* rigorous and practically relevant—that it can engage with abstract philosophical questions about the nature of artificial and natural minds while also addressing ethical challenges posed by real-world technologies. These chapters aim not only to explore the performance of AI models and their use in science and society but also to highlight the importance of designing philosophically-grounded experiments that look under the hood of the models—a principle deeply rooted in empirical work in computer science—to understand what they can teach us about ourselves. The truth is that while philosophical work on AI aims to balance philosophical inquiry with empirical findings, in practice, things often end up lopsided—and you can guess which way. A healthy dose of hands-on exploration under the hood of AI models might just do us, philosophers, good. At the same time, a healthy dose of engagement with philosophical inquiry might just do computer scientists, and AI research more generally, good.